Tzu-Chieh (Jeremy) Chao

+1-352-219-7301 | jeremy880524@gmail.com | tzuchiehchao | GitHub | Personal Website

EDUCATION

University of Florida **Aug 2024 - May 2026**
Master of Science, Applied Data Science *Gainesville, FL*
- **Achievements:** Achievement Award Scholarship

National Taipei University **Sep 2017 - Jan 2022**
Bachelor of Business Administration, Business Administration *Taipei, Taiwan*
- **Coursework:** Linear Algebra, Probability and Statistics, Regression Analysis, Business Analytics

WORK EXPERIENCE

Aventusoft LLC **Aug 2025 - May 2026**
Machine Learning Engineer – UF IPPD Co-op *Gainesville & Boca Raton, FL*
- Collaborated with Aventusoft engineers via Git, Azure DevOps, and CI/CD pipelines within cloud environments to define system requirements, software architecture, and deployment constraints for a clinical-grade ECG AI system.
- Designed cloud and mobile inference pipelines using ONNX Runtime, CoreML, and TFLite, targeting <50 MB model size and <30 s on-device latency.
- Supported FDA Design Control documentation aligned with FDA-oriented medical software development practices.
- Led cross-functional communication with sponsor liaisons to review validation strategy, dataset alignment, and technical deliverables across the project lifecycle.

Commerce Development Research Institution **Sep 2022 - Jul 2023**
Business Analyst Consultant *Taipei, Taiwan*
- Analyzed sustainability and ESG disclosures from 30 Fortune 500 companies using pandas and scikit-learn to identify net-zero implementation patterns and business implications relevant to Taiwanese industries.
- Conducted data-driven operational impact analysis and developed data models to support competitiveness and strategy assessments for SMEs.
- Built ETL processes using SQL, dbt, and pandas to integrate sustainability data into a Snowflake analytical data warehouse on Azure for exploratory analysis and visualization.
- Collaborated with senior research scientists to translate analytical findings into phased, data-backed recommendations aligned with Taiwan’s 2050 net-zero emissions roadmap.

SELECTED PROJECTS

Electrocardiogram Deep Learning **Aug 2025 - May 2026**
University of Florida & Aventusoft, LLC
- Built end-to-end preprocessing pipelines in Python using Pandas and NumPy for single-lead 500 Hz ECG data, including bandpass filtering, normalization, sliding-window segmentation, and harmonization across LUDB, QTDB, PTB-XL, and CPSC datasets in an Azure cloud environment
- Implemented a 1D U-Net segmentation model in PyTorch with Gaussian soft labels and morphology-aware post-processing in TensorFlow, achieving 1.5–6 ms fiducial timing error for P/QRS/T delineation
- Designed and evaluated arrhythmia classification models using a Transformer-based ECG-FM foundation model and a lightweight CNN-BiLSTM classifier for mobile deployment.
- Conducted benchmarking, robustness testing, model pruning, and quantization using ONNX Runtime and TFLite to optimize latency for cloud (<2 min) and mobile (<30 s) real-time inference pipelines with Docker and CI/CD integration
- Collaborated with Aventusoft engineers to define system requirements and deployment constraints.

TECHNICAL SKILLS

- **Programming**: Python, C++, C, SQL, JavaScript, HTML, CSS
- **Statistical Tools**: R, SPSS, Excel VBA
- **Data Visualization**: Power BI, Tableau
- **Machine Learning & Data Science**: PyTorch, TensorFlow, Dbt, Snowflake, Databricks, Scikit-learn, Pandas
- **MLOps, Deployment & Cloud**: ONNX Runtime, CoreML, TFLite, FastAPI, Docker, Flask, MLOps, Git, CI/CD, APIs, Unix, Linux, Windows, MacOS, Cloud Environments, Azure, Amazon S3